

Review

2.6 Syllabus covered

- ✓ Binary numbers
- ✓ Generate minterms, maxterms, SOP canonical form and POS canonical forms and convert between them
- ✓ Understand and use the laws and theorems of Boolean Algebra
- ✓ Perform algebraic simplification using Boolean algebra
- ✓ Simplification using K-maps
- ✓ Derive sum of product and product of sums expressions for a combinational circuit
- ✓ Convert combinational logic to NAND-NAND and NOR-NOR forms
- Hexadecimal, Sign-magnitude, One's-complement and Two's complement. Conversions between them.
- Design combinational circuits for positive and negative logic
- Design Hazard-free two level circuits and understand Hazards in multi-level circuits
- Compute noise margin of one device
- Describe how tri-state and open-collector outputs are different from totem-pole outputs.
- Different between and limitations of master-slave and edge-triggered flip-flops.
- Compute fan out and noise margin of one device driving the same time
- Know the differences and similarities between PAL, PLA, and ROMs and can use each for logic design
- Design combinational circuits using multiplexers and decoders
- Analyze a sequential circuit and derive a state-table and a state-graph
- Understand the difference between synchronous and asynchronous inputs
- Derive a state graph or state table from a word description of the problem
- Reduce the number of states in a state table using row reduction and implication tables
- Perform a state assignment using the guideline method

- Implement a design using JK, SR, D or T flip-flops
- Analyse and design both Mealy and Moore sequential circuits with multiple inputs and multiple outputs
- Convert between Mealy and Moore designs
- Partition a system into multiple state machines

2.6.1 Labs (not questioned in exams)

- Use computer tools to enter designs graphically and HDL
- Simulate designs using computer tools
- Use computer tools to program gate arrays logic and debug and test

Chapter 3

Sample midterm exam

Student Name:

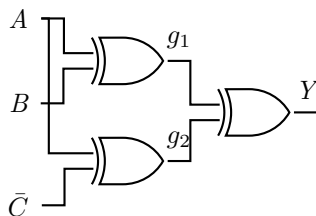
Student Email:

3.1 Instructions

75 min

- Time allowed is ~~50~~ minutes. (This sample exam might be lengthier than the actual exam. Same instruction will apply to the midterm.)
- In order to minimize distraction to your fellow students, you may not leave during the last 10 minutes of the examination.
- The examination is closed-book. One 8x11in cheatsheet is allowed.
- Non-programmable calculators are permitted.
- The maximum number of marks is 80, as indicated; the midterm examination amounts 10% toward the final grade.
- Please colored pen/pencils for K-maps, use a pen or heavy pencil to ensure legibility.
- Please show your work; where appropriate, marks will be awarded for proper and well-reasoned explanations.

Problem 3.1. Consider the circuit below



$$g_1 = A \oplus B$$

$$g_1 = m_1 + m_2 = \bar{A}B + A\bar{B}$$

$$g_2 = \bar{A}\bar{C} + A\bar{C} = \bar{A}\bar{C} + AC$$

A	B	g_1
0	0	0
0	1	1
1	0	1
1	1	0

← m_1
← m_2

By algebraic manipulation, prove or disprove that $Y = \bar{B}\bar{C} + BC$ (10 marks).

Problem 3.2. Use the following 4-variable K-map for $F(A, B, C, D)$, and find a minimal sum of products (SOP) expression for F (15 marks)

$$\overline{(\bar{A}\bar{C} + AC)} \text{ XOR}$$

$$= (A + C)(\bar{A} + \bar{C})$$

$$= A\bar{A} + C\bar{C} + A\bar{C} + \bar{A}C$$

$$= 0 + 0 + A\bar{C} + \bar{A}C$$

$$= A\bar{C} + \bar{A}C \leftarrow \text{XOR}$$

$$Y = g_1 \bar{g}_2 + \bar{g}_1 g_2$$

$$= \underbrace{(\bar{A}B + A\bar{B})}_{\text{XOR}} \underbrace{(\bar{A}\bar{C} + AC)}_{\text{XOR}}$$

$$+ (\bar{A}B + A\bar{B})(\bar{A}\bar{C} + AC)$$

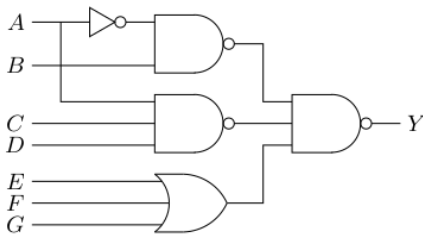
$$= (\bar{A}B + A\bar{B})(A + C)(\bar{A} + \bar{C})$$

$$\begin{aligned}\overline{(\bar{A}B + A\bar{B})} &= (A + \bar{B})(\bar{A} + B) \\ &= A\bar{A} + \bar{B}B + A\bar{B} + A\bar{B} \\ &= AB + \bar{A}\bar{B}\end{aligned}$$

$$\begin{aligned}Y &= (\bar{A}B + A\bar{B})(\bar{A}\bar{C} + \bar{A}C) \\ &\quad + (\bar{A}B + A\bar{B})(\bar{A}\bar{C} + AC) \\ &= \cancel{A\bar{B}}\bar{B}\bar{C}^0 + \bar{A}B\bar{C} + A\bar{B}\bar{C}^0 \\ &\quad + A\bar{B}C + \bar{A}\bar{B}\bar{C} + \cancel{A\bar{B}}\bar{C}^0 \\ &= (A + \bar{A})B\bar{C} + (\bar{A} + A)\bar{B}\bar{C} \\ &= B\bar{C} + \bar{B}\bar{C}\end{aligned}$$

	AB			
	00	01	11	10
CD				
00	1	0	0	1
01	1	1	0	1
11	0	1	0	0
10	0	1	1	0

Problem 3.3. Use boolean algebra to find an SOP expression for Y in the circuit below. Draw an equivalent circuit beside the given circuit (5 marks).



Problem 3.4. Consider the function Y given below.

$$Y(A, B, C, D) = \sum m(0, 3, 5, 7, 8, 14) + d(2, 12, 15)$$

1. Draw a K-maps to derive a minimum SOP and POS expressions for Y . Indicate all essential prime implicants for Y or $\bar{Y}$ in your K-maps (20 marks).
2. Sketch a two-level NOR-NOR circuit for Y . Assume that A , B , C , and D are available in true and complementary forms (5 marks).
3. Write Y in Product of sums (POS) canonical form (5 marks).

Problem 3.5. Design a minimal SOP circuit to add two two-bit unsigned numbers. Denote the two bits of first number as A_1A_0 and the two bits of second number as B_1B_0 . The result will be a 2-bit sum S_1S_0 and a carry C . Start with filling out the following truth table (3 example rows are provided) and then use K-maps to find minimal SOP for S_1 , S_0 and a single carry bit C_1 (20 marks).

A_1A_0	00	01	10	11
B_1B_0	00	01	10	11

$$\begin{array}{r}
 A_1A_0 \\
 + B_1B_0 \\
 \hline
 C \quad S_1S_0
 \end{array}$$

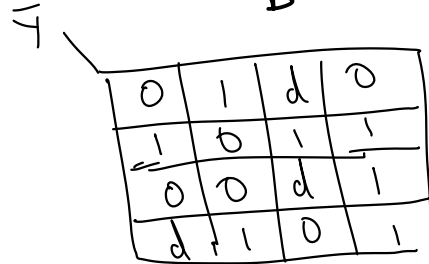
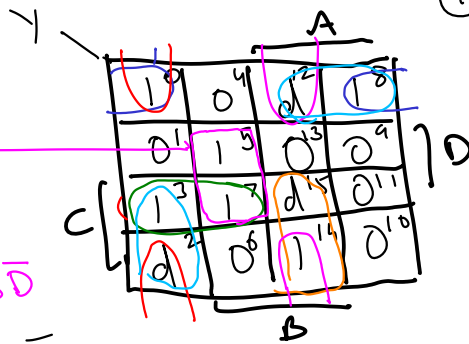
Problem 3.4. Consider the function Y given below.

$$Y(\overset{\text{MSB}}{\underline{A}}, \overset{\text{LSB}}{\underline{B}}, C, D) = \sum m(0, 3, 5, 7, 8, 14) + d(2, 12, 15)$$

1. Draw a K-map to derive a minimum SOP and POS expressions for Y . Indicate all essential prime implicants for Y or $\bar{Y}$ in your K-maps (20 marks).
2. Sketch a two-level NOR-NOR circuit for Y . Assume that A , B , C , and D are available in true and complimentary forms (5 marks).
3. Write Y in Product of sums (POS) canonical form (5 marks).

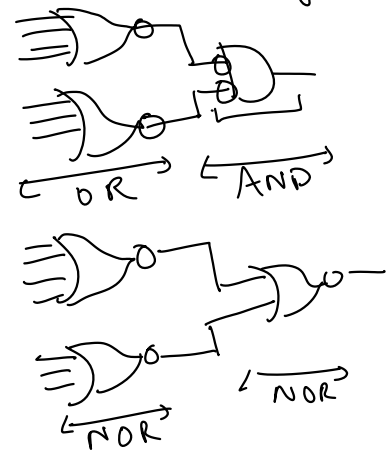
EPI
= $\bar{A}BD$

$$Y = \bar{A}BD + \bar{A}\bar{B}C + \bar{B}C\bar{D} + AB\bar{D}$$

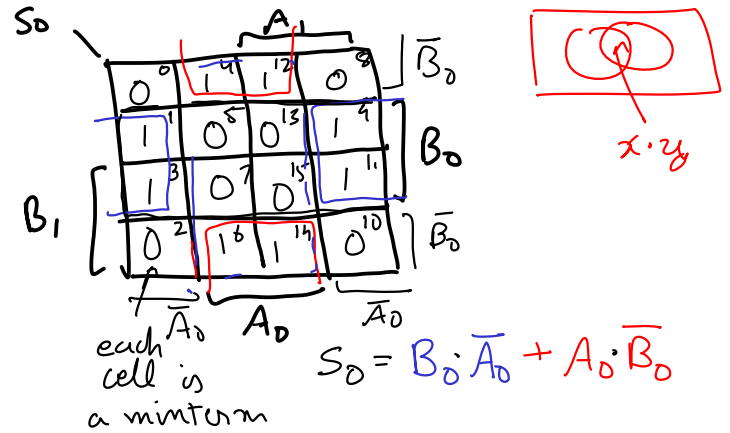


- ① The process of grouping:
- Finding all Prime implicants
- ② Finding the minimal cover
(finding fewest PI that cover all 1's)

Product of sums form



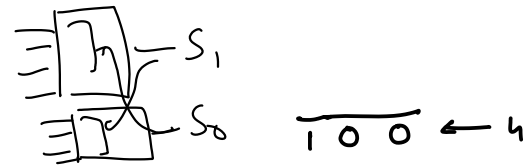
msb	A_1	A_0	B_1	B_0	C_1	S_1	S_0
	0	0	0	0	0	0	0
	0	0	0	1	0	0	1
	0	0	1	0	0	1	0
	0	0	1	1	0	1	1
	0	1	0	0	0	0	1
	0	1	0	1	0	1	0
	0	1	1	0	0	1	1
	0	1	1	1	1	0	0
	1	0	0	0	0	1	0
	1	0	0	1	1	0	1
	1	0	1	1	1	0	1
	1	1	0	0	1	1	1
	1	1	0	1	1	0	0
	1	1	1	0	1	1	1
	1	1	1	1	1	1	0



Restricted

- 4-5 inputs
- two level circuits
 - Sum of products
 - Product of sums
- single output

Quine McCluskey

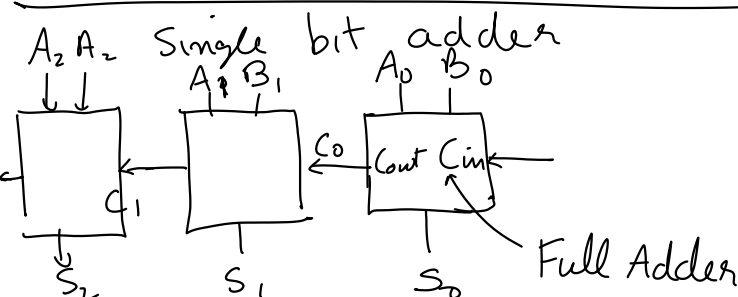


$A_1 A_0$	0	0
$B_1 B_0$	0	0
	0	0
C_1, S_1, S_0		

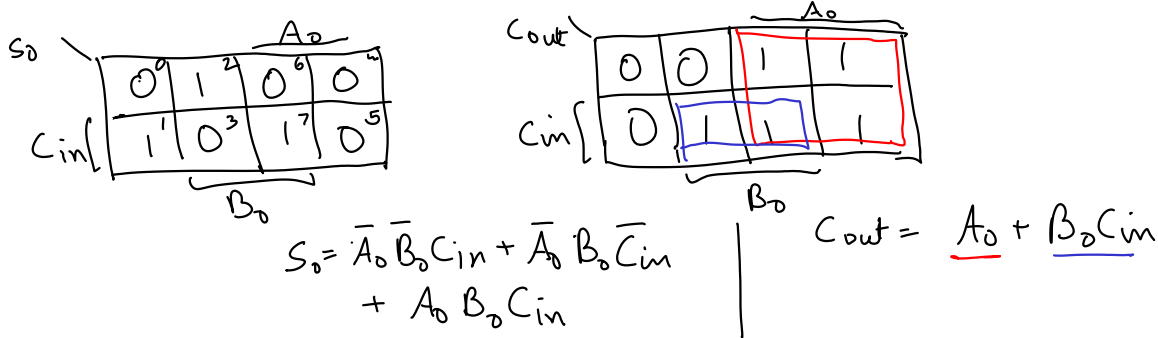
1
0 1
0 1
0 1 0

0 1
1 0
1 1

1
0 1
1 1



A_0	B_0	C_{in}	C_{out}	S_0
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



Chapter 4

Muxes and Decoders

Multiplexers

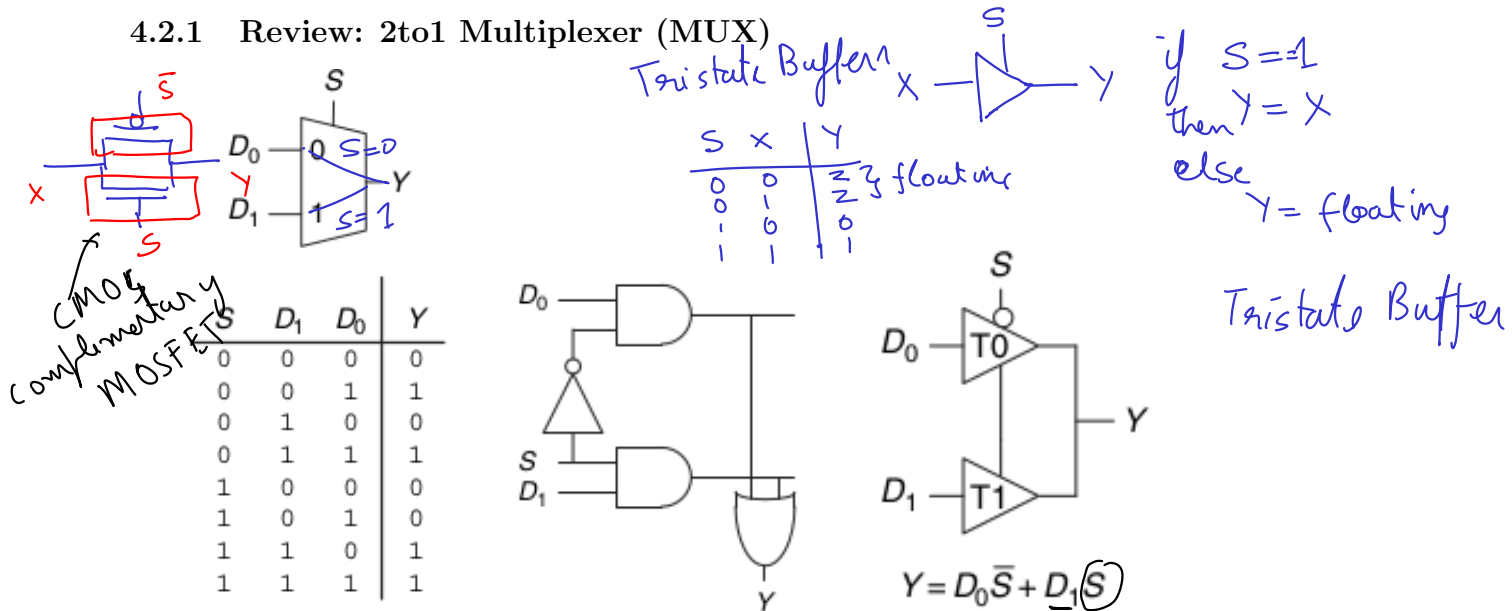
Building blocks of combinational circuits that you see repeatedly.

4.1 Objectives

1. Design combinational circuits using multiplexers and decoders

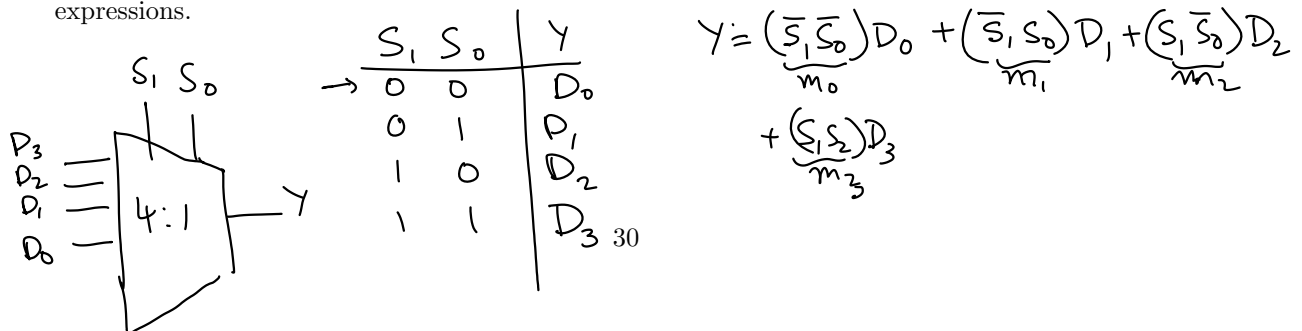
4.2 Design combinational circuit using multiplexers [1, Section 2.8.1]

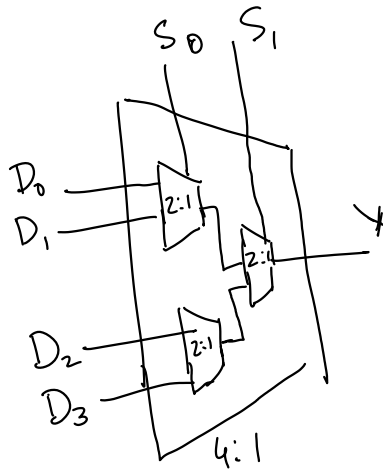
4.2.1 Review: 2to1 Multiplexer (MUX)



4.2.2 Wider multiplexers

Draw the symbol for a 4:1 MUX, an 8:1 MUX and a $2^N : 1$ MUX and write corresponding Boolean expressions.





$y (S_1 S_0 = 00)$

,

.

:

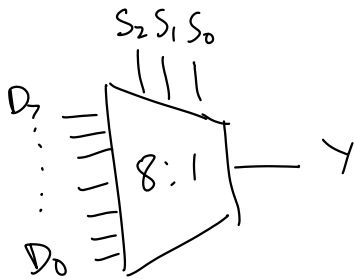
.

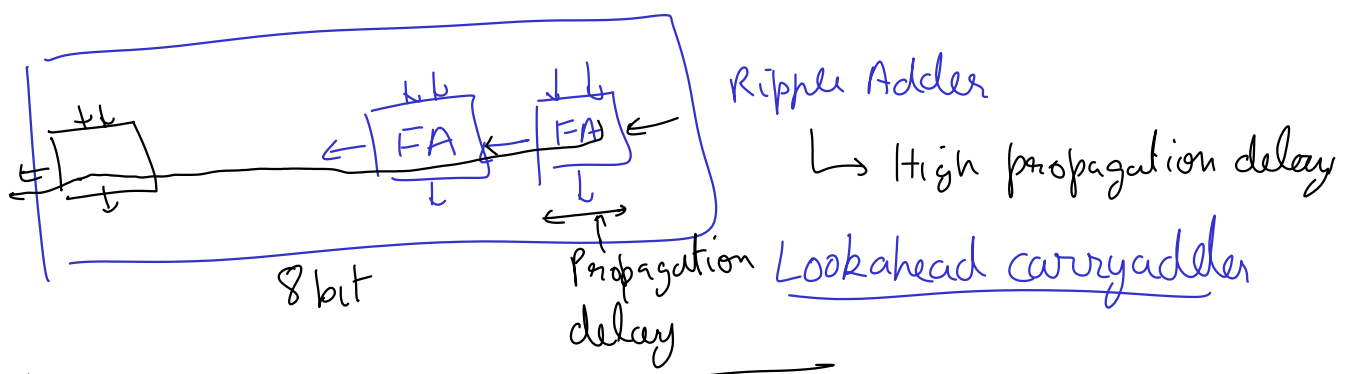
```

if (S1 == 0) {
    if (S0 == 0) {
        Y = D0;
    } else {
        Y = D1;
    }
} else {
    if (S0 == 0) {
        Y = D2;
    } else {
        Y = D3;
    }
}

```

8:1 MUX



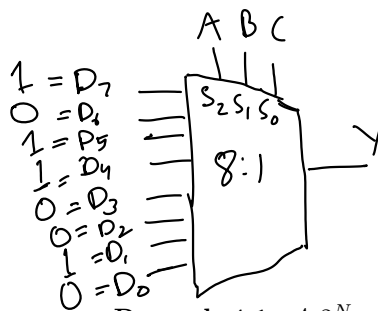


circuit timing

↳ 8 x Propagation delay

↳ Contamination delay

3 inputs

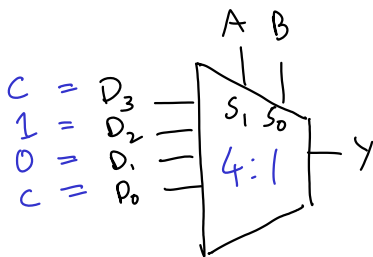
 3 selection bits ³¹
Example 4.1. Design a circuit for $Y = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}BC$ using a 8:1 MUX.


A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

A			
0	0	1	1
0	1	0	1
C	0	0	1
B	1	0	1

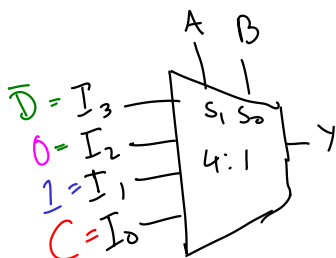
$$Y = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}BC$$

Remark 4.1. A $2^N : 1$ MUX can be used to program any N -input logic function.

Example 4.2. Design a circuit for $Y = \bar{A}\bar{B} + \bar{B}\bar{C} + \bar{A}BC$ using a 4:1 MUX and NOT gates only.


2 selection bits			
$\bar{A}\bar{B}$	$\bar{A}B$	$AB\bar{A}$	$A\bar{B}$
0	0	0	1
1	0	1	1
C	0	1	1
D0	1	0	1
D1	0	1	1
D2	0	1	1
D3	0	1	1

Remark 4.2. A $2^{N-1} : 1$ MUX can be used to program any N -input logic function, if we use literals on the input side.

Example 4.3. Design a circuit for $Y = \bar{A}\bar{C} + \bar{A}B + B\bar{D}$ using a 8:1 MUX and NOT gates only. Also design using 4:1 MUX and other gates. fewest gates.


Repeat this process with AC connect to selection bits, BD connected to Input lines

AC			
$\bar{A}\bar{C}$	$\bar{A}C$	$AC\bar{B}$	ACB
0	1	1	0
0	1	0	0
1	1	0	0
1	1	1	0
C	1	1	1
B	1	1	1
D	1	1	1

$$Y = (\bar{A}\bar{B})I_0 + (\bar{A}B)I_1 + (A\bar{B})I_2 + (AB)I_3$$

Problem 4.1. (15 marks [1, Ex-2.42]) Implement the function $Y = BC + \bar{A}\bar{B}\bar{C} + B\bar{C}$ using

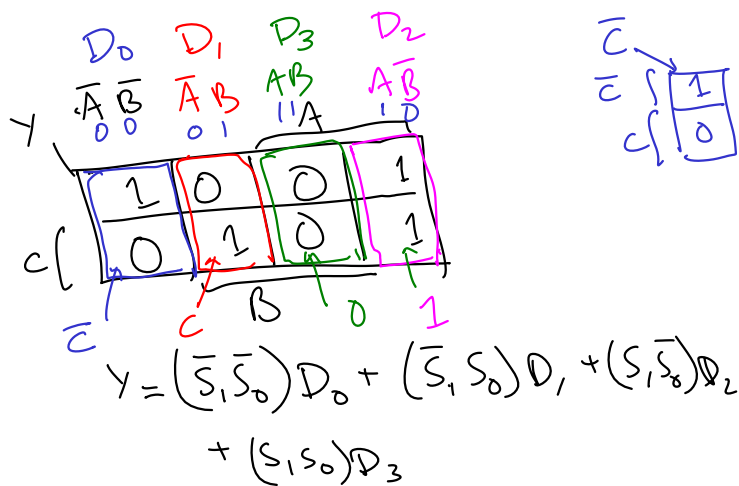
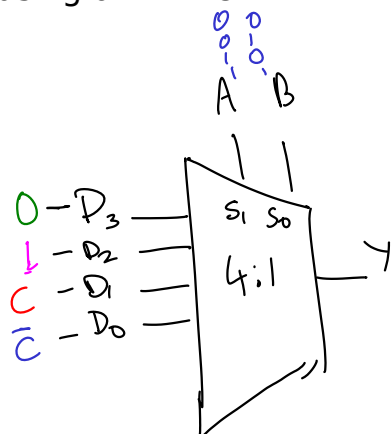
1. an 8:1 multiplexer
2. a 4:1 multiplexer and no other gates
3. a 2:1 multiplexer, one OR gate, and an inverter

4.3 Encoders and Decoders

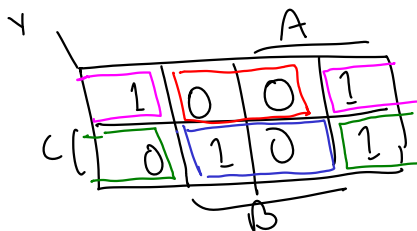
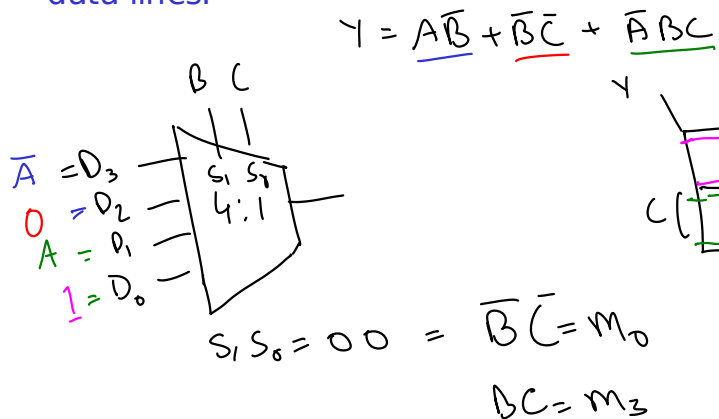
Example 4.4. Draw the symbol and the truth table for 2:4 decoder. Also write the logic expressions.

$$Y = \underline{A\bar{B}} + \underline{\bar{B}\bar{C}} + \underline{\bar{A}BC}$$

Design a circuit for this expression using a 4:1 MUX



Repeat this design process with BC as the selection bits, with A connected to data lines.



$$Y = (BC)D_3 + (B\bar{C})D_2 + (\bar{B}C)D_1 + (\bar{B}\bar{C})D_0$$

Example 4.5. Draw the symbol and the truth table for 3:8 decoder, 4:16 decoder and $N : 2^N$ decoder. Also write the logic expressions.

Example 4.6. Design a circuit for a XOR gate using a 2:4 decoder and an OR gate.

Example 4.7. Design a circuit for $Y = A\bar{B} + \bar{B}\bar{C} + \bar{A}BC$ using a 3:8 decoder and an OR gate.

4.3.1 (Priority) Encoders

Example 4.8. Draw symbol and truth table for a 4:2 priority encoder.

Example 4.9. Draw symbol and truth table for a 8:3 priority encoder.